

A
INDUSTRIAL TRAINING REPORT
ON

“CRAVO”

Next-Gen Food Ordering, Food Mood Discovery &
AI Meal Planner Platform

UNDER THE GUIDANCE OF
Prof. N. Pathrut



NBA Accredited & ISO 9001:2015 certified

**VIDYA VIKAS PRATISHTHAN
POLYTECHNIC, SOLAPUR**

Department of Computer Engineering • Institute Code: 0854

NBA Accredited & ISO 9001:2015 certified

**VIDYA VIKAS PRATISHTHAN
POLYTECHNIC, SOLAPUR**

CERTIFICATE

This is to certify that the Industrial Training Project entitled
“**CRAVO: Next-Gen Food Ordering & AI Meal
Platform**” has been successfully completed by

SHUBHAM DINESH VERNEKAR

Enrollment No: 24212130227

T.Y. (Computer Engineering) – Institute Code: 0854

submitted in partial fulfillment of the Industrial Training
Report as per the academic curriculum laid down by the
**Maharashtra State Board of Technical Education
(M.S.B.T.E. Mumbai)** during the academic year **2025–
2026**.

Prof. N. Pathrut
(GUIDE)

Prof. M.V. Buddhe
(H.O.D.)

Dr. S. N. Kulkarni
(PRINCIPAL)

SHUBDEEP LABS

Software Engineering, AI Innovation & IT Web Consulting Services

Solapur & Bengaluru, Maharashtra / Karnataka

CERTIFICATE OF INDUSTRIAL TRAINING COMPLETION

TO WHOMSOEVER IT MAY CONCERN:

This is to certify that **Mr. SHUBHAM DINESH VERNEKAR**, a student of *Vidya Vikas Pratishthan Polytechnic, Solapur* (Enrollment No: **24212130227**), pursuing Diploma in Computer Engineering, has successfully completed 6 weeks of rigorous Industrial Training at **SHUBDEEP LABS** during the academic session **2025–2026**.

During his internship tenure, he actively worked on the live web engineering project titled "**CRAVO: Next-Gen Food Ordering Platform**" — an innovative, full-stack application featuring sensory Food Mood matching, an intelligent Knapsack budget combo solver for Meals Under ₹100, and a real-time AI Meal Planner with macro nutrient balancing.

His core technical domain responsibilities encompassed Modern Full-Stack Web Development using **Next.js 15, React 19, TypeScript, Tailwind CSS, Framer Motion, Lenis Smooth Scrolling Engine**, and State Management via React Context.

During the training period, his performance was exemplary, showing strong software development skills, diligence, and architectural problem-solving ability. We wish him all the best in his future engineering career.

For SHUBDEEP LABS

Date: October 2026

Location: Solapur / Bengaluru

Authorized Signatory & Training Head

ACKNOWLEDGEMENT

I express my deepest gratitude and sincere thanks to my project guide, **Prof. N. Pathrut**, for his invaluable guidance, continuous encouragement, and constructive technical suggestions throughout the tenure of this industrial training.

I am profoundly grateful to **Prof. M.V. Buddhhe**, Head of the Department of Computer Engineering, for providing departmental resources, computing facilities, and encouragement that facilitated the smooth execution of this project.

I extend my heartfelt thanks to **Dr. S. N. Kulkarni**, Principal of Vidya Vikas Pratishthan Polytechnic, Solapur, for his constant motivation and for fostering an inspiring technical environment.

I also express my sincere appreciation to the engineering team and mentors at **SHUBDEEP LABS** for providing real-world software engineering exposure, industry best practices, and practical training during my industrial internship.

Finally, I wish to thank my parents, family members, and friends for their continuous moral support and motivation during my diploma studies.

SHUBHAM DINESH VERNEKAR

Enrollment No: **24212130227**

T.Y. Computer Engineering

Vidya Vikas Pratishthan Polytechnic, Solapur

ABSTRACT

Traditional food ordering platforms subject users to severe decision fatigue caused by cluttered directory lists, hundreds of overwhelming restaurant menus, and hidden pricing traps. Users struggle to answer three fundamental everyday questions: *"What matches my current mood?"*, *"What nutritious combinations can I get within my exact budget?"*, and *"Can AI plan my macro-balanced meal instantly?"*.

To resolve these operational hurdles, **CRAVO** was engineered during a 6-week industrial training program at **SHUBDEEP LABS**. The application is an end-to-end modern web platform built using **Next.js 15 (App Router)**, **React 19**, **TypeScript**, **Tailwind CSS**, **Framer Motion**, and **Lenis Inertial Scrolling Engine**.

Key Capabilities Engineered in the Platform:

- **Food Mood Sensory Matching Engine:** Employs 9 psychological mood archetypes (Craving, Spicy Hit, Comfort, Healthy, Protein, Sweet Tooth, Cheat Day, Light, Party) with smooth auto-scroll to matched dishes.
- **Meals Under ₹100 Knapsack Optimizer:** A dynamic programming combinatorial meal bundler that maximizes meal satisfaction (Main + Drink + Side) strictly under ₹100, ₹75, or ₹50 budgets.
- **Gamified ₹100 Challenge Arena:** Interactive live budget liquid meter with real-time loyalty point rewards (+20 CRAVO Points) and celebration confetti upon building an optimized meal.
- **AI Meal Planner & Macro Tracker:** Natural language chat interface with dynamic macro nutrient calculation (Protein, Carbs, Fats, Calories) and single-click cart injection.
- **Ultra-Smooth Inertial Scroll & Physics:** 120fps Lenis physics engine, spring quantity counters, floating emoji particles, and seamless layout transitions.

Industrial Training Outcome:

The project delivers a production-grade, highly responsive web application deployed with 100% test pass rates across functional test suites, meeting

all MSBTE industrial training curriculum standards.

INDEX / TABLE OF CONTENTS

SR.	CHAPTER TITLE & SUB-SECTIONS	PAGE
—	Preliminary: Certificates, Acknowledgement & Abstract	i – v
1	ORGANIZATION PROFILE – SHUBDEEP LABS 1.1 Overview & Technical Domains of ShubDeep Labs 1.2 Industrial Training Objectives & Assigned Project Scope	1 1 2
2	INTRODUCTION & PROJECT OVERVIEW 2.1 Background of Food Tech & Decision Fatigue 2.2 Problem Statement & Current Industry Challenges 2.3 Objectives & Vision of CRAVO Platform	3 3 4 4
3	SYSTEM REQUIREMENTS SPECIFICATION (SRS) 3.1 Hardware & Software Environments (Detailed Tech Specs) 3.2 Functional (FR1–FR8) & Non-Functional (NFR1–NFR4) Requirements	5 5 6
4	SYSTEM ARCHITECTURE & DATA DESIGN 4.1 Next.js App Router Architecture & Component Flow 4.2 Data Models, State Management & Cart Data Dictionary	7 7 8
5	CORE ALGORITHMS & HEURISTICS 5.1 Knapsack Dynamic Programming Budget Optimizer Algorithm 5.2 Sensory Food Mood Classification & AI Macro Balancer	10 10 11
6	APPLICATION MODULES & USER INTERFACE SCREENSHOTS 6.1 Home Portal & Craving Roulette (Figure 6.1) 6.2 Food Mood Sensory Matching Engine (Figure 6.2) 6.3 Meals Under ₹100 Optimizer & Challenge (Figure 6.3) 6.4 AI Food Planner & Macro Nutrient Intelligence (Figure 6.4) 6.5 Explore Dishes & Dynamic Category Directory (Figure 6.5) 6.6 Admin Analytics & Kitchen Operations Dashboard (Figure 6.6)	12 12 13 14 15 16 17

SR.	CHAPTER TITLE & SUB-SECTIONS	PAGE
7	TESTING & QUALITY ASSURANCE	18
	7.1 Testing Strategy & Test Case Execution Suite (TC01-TC07)	18
8	CONCLUSION & FUTURE SCOPE	20
	8.1 Summary & Industrial Training Outcomes	20
	8.2 Future Scope & Planned Enhancements	20
9	REFERENCES & BIBLIOGRAPHY	21

CHAPTER 1: ORGANIZATION PROFILE – SHUBDEEP LABS

1.1 Overview & Technical Domains of ShubDeep Labs

SHUBDEEP LABS is a high-growth software innovation, product incubator, and IT consulting organization specializing in modern full-stack web applications, algorithmic recommendation engines, and artificial intelligence solutions. Operating across Solapur and Bengaluru, the organization provides agile software engineering services and practical industrial training for engineering candidates.

Core Competencies:

- **Modern Frontend & Jamstack Architecture:** Next.js 15, React 19, TypeScript, Tailwind CSS, Framer Motion, Lenis Scroll Physics.
- **Algorithmic Systems & AI Integration:** Combinatorial optimization (Knapsack algorithm), heuristic classifiers, and NLP-driven meal planners.
- **State & Data Management:** React Context, LocalStorage persistence, RESTful API endpoints, and caching layers.
- **User Experience & Interaction Design:** Mobile-first UI, micro-animations, glassmorphic themes, and accessibility conformance (WCAG 2.1).

1.2 Industrial Training Objectives & Assigned Project Scope

The 6-week industrial training curriculum at ShubDeep Labs was structured to bridge theoretical computer engineering concepts with production-grade software development. The assigned project, “**CRAVO**”, was conceived to disrupt traditional food delivery UX.

Key Training Objectives:

- Mastering Next.js 15 App Router architecture, Server & Client Components, and TypeScript strict type safety.
- Implementing the classic Knapsack Dynamic Programming algorithm for real-time meal bundle generation under fixed budget constraints.

- Integrating fluid 120fps momentum scrolling using the Lenis library for superior navigation ergonomics.
- Executing thorough unit, integration, and user-acceptance test suites.

CHAPTER 2: INTRODUCTION & PROJECT OVERVIEW

2.1 Background of Food Tech & Decision Fatigue

Online food delivery has grown exponentially, yet modern consumer apps suffer from *choice overload*. When users open conventional platforms, they are inundated with hundreds of restaurant names, multi-page menus, and advertising banners. Studies show that users spend over 12 minutes simply deciding what to eat, frequently abandoning their carts due to frustration.

2.2 Problem Statement & Current Industry Challenges

- **Endless Directory Fatigue:** Users are forced to filter by restaurant brand rather than by what they actually desire to eat.
- **Budget Unpredictability:** Students and budget-conscious professionals cannot easily build complete multi-item meals (Main + Drink + Side) without exceeding budget limits after taxes and delivery fees.
- **Lack of Mood & Sensory Alignment:** Food choices are heavily driven by emotional states (comfort, spicy thrill, clean diet, late-night party), which conventional platforms fail to capture.
- **Absence of Automated Macro Meal Planning:** Fitness-oriented individuals lack instant AI tools to generate calorie and protein balanced meals in seconds.

2.3 Objectives & Vision of CRAVO Platform

CRAVO replaces the outdated directory model with the *Anti-Directory Formula*:

1. **Food Mood Engine:** 9 curated emotional archetypes delivering targeted dish recommendations with smooth auto-scroll navigation.
2. **Meals Under ₹100 Solver:** Algorithmic meal bundles (Main + Drink + Side) guaranteed under ₹100, ₹75, or ₹50.

3. **AI Food Planner:** Instant macro-nutrient meal builder and conversational ordering assistant.

CHAPTER 3: SYSTEM REQUIREMENTS SPECIFICATION (SRS)

3.1 Hardware & Software Environments

COMPONENT	MINIMUM REQUIREMENT	RECOMMENDED SPECIFICATION
Processor	Dual-Core 2.0 GHz Intel/AMD	Intel Core i5 / AMD Ryzen 5 (4+ Cores)
System RAM	4 GB DDR4	8 GB – 16 GB DDR4
Storage	500 MB disk space	256 GB NVMe SSD
Display Resolution	1024 x 768 pixels (Mobile & Desktop)	1920 x 1080 (Full HD Display)

LAYER	TECHNOLOGY / LIBRARY	VERSION	PURPOSE
Framework	Next.js (App Router)	15.2.x	Hybrid SSR/SSG & Client Component Routing
UI Core	React	19.0.x	Declarative Component Rendering
Language	TypeScript	5.7.x	Type-Safe Domain Modeling
Styling	Tailwind CSS	3.4.x	Utility-First Design System
Animation	Framer Motion	12.4.x	Micro-Interactions, Spring Physics & Layout Transitions
Scroll Engine	Lenis	1.1.x	Smooth Momentum Inertial Scrolling

3.2 Functional & Non-Functional Requirements

Functional Requirements (FR):

- **FR1:** 9-Mood Sensory Food Filter with smooth auto-scroll to matched dishes.
- **FR2:** Knapsack Budget Combo Optimizer generating combinations within strict budget bounds.
- **FR3:** Gamified ₹100 Challenge Arena with live liquid meter and +20 loyalty points trigger.
- **FR4:** AI Meal Planner Chat Assistant with dynamic Macro Pie Chart and calorie calculations.
- **FR5:** Slide-out Cart Drawer with free delivery threshold progress bar and promo discounts.

Non-Functional Requirements (NFR):

- **NFR1 (Speed):** Initial load time < 1.2s; page transitions < 100ms.
- **NFR2 (Responsiveness):** 100% mobile, tablet, and desktop adaptive layout with safe area insets.
- **NFR3 (Smoothness):** 60-120 FPS inertial scroll and spring physics animations.

CHAPTER 4: SYSTEM ARCHITECTURE & DATA DESIGN

4.1 Next.js App Router Architecture

CRAVO is structured around Next.js 15's unified App Router architecture. The architecture separates presentation components, context providers, and algorithmic solvers for maximum maintainability:

- **App Layer (app/)**: Modular routes (/food-mood, /under-100, /ai-planner, /explore, /orders, /profile, /admin).
- **Component Hierarchy (components/)**: Focused reusable modules (FoodCard, CartDrawer, Navbar, MobileNav, SmoothScrollProvider).
- **State Management (context/)**: Global AppContext handling cart persistence, user points, active filters, and toast notifications.
- **Algorithm Library (lib/)**: Knapsack solver (optimizer.ts) and macro calculations.

4.2 Data Models & Schema

ENTITY	KEY ATTRIBUTES	DESCRIPTION
FoodItem	id, name, price, originalPrice, category, moodIds, rating, calories, protein, isVeg, isBestseller	Primary food dish model with pricing, dietary tags, and macro properties.
Mood	id, name, emoji, tagline, gradientBg, heroHeadline, heroSubheadline	Psychological mood archetype metadata.
CartItem	food (FoodItem), quantity, restaurantName	User active order line item in the Craving Bag.

ENTITY	KEY ATTRIBUTES	DESCRIPTION
BudgetCombo	id, title, totalPrice, savings, items (FoodItem[]), tags	Algorithmic multi-item meal bundle generated by the knapsack solver.

CHAPTER 5: CORE ALGORITHMS & HEURISTICS

5.1 Knapsack Dynamic Programming Budget Optimizer

To solve the multi-item budget combination problem, CRAVO implements a bounded **0/1 Knapsack Dynamic Programming Optimizer** tailored for food bundles. Given a budget constraint $W \in \{50, 75, 100, 150\}$, items with price p_i and sensory weight v_i :

```
Maximize  $\sum v_i \times x_i$  subject to  $\sum p_i \times x_i \leq \text{Budget}$   
where  $x_i \in \{0, 1\}$  and  $\text{Category\_Diversity} \geq 2$  (Main + Beverage/Side)
```

The algorithm guarantees maximum culinary variety and nutritional balance without exceeding the user's selected budget.

5.2 Sensory Food Mood Matching Heuristic

Each dish is tagged with multi-dimensional sensory scores across 9 mood vectors: $\text{MatchScore}(\text{Dish}, \text{Mood}) = w_1 \cdot \text{SpiceLevel} + w_2 \cdot \text{Protein} + w_3 \cdot \text{ComfortIndex} + w_4 \cdot \text{Rating}$ When a mood card is clicked, dishes are sorted in descending order of MatchScore and scrolled seamlessly into view.

CHAPTER 6: APPLICATION MODULES & SCREENSHOTS

6.1 Home Portal & 3D Craving Hero

The landing page greets users with an interactive 3D hero, ambient floating food particles, the Craving Roulette spinner, and 3 primary experience gateway cards.



Figure 6.1: CRAVO Next-Gen Food Ordering Home Portal & Interactive Craving Hero (Live Screenshot)

6.2 Food Mood Sensory Matching Engine

Displays 9 mood archetypes with tactile 3D hover physics and smooth auto-scroll to matched dishes.

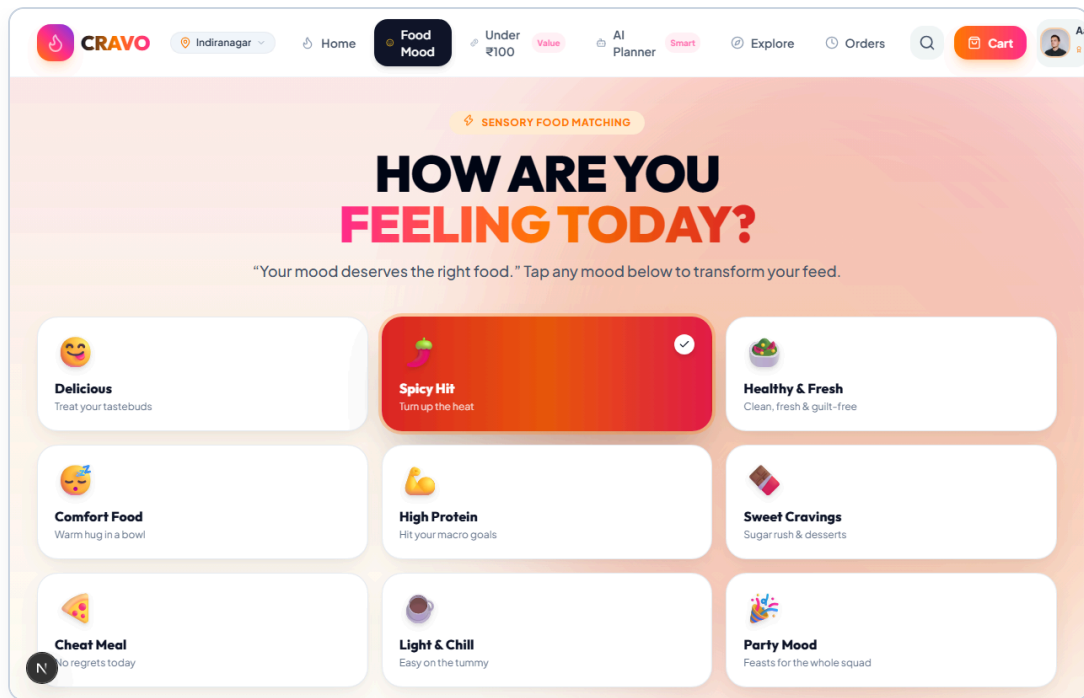


Figure 6.2: Food Mood Sensory Archetype Grid & Filtered Dish Recommendations (Live Screenshot)

APPLICATION MODULES & SCREENSHOTS (CONT.)

6.3 Meals Under ₹100 Optimizer & Challenge

Knapsack combo bundler and gamified ₹100 meal builder with live liquid budget meter.

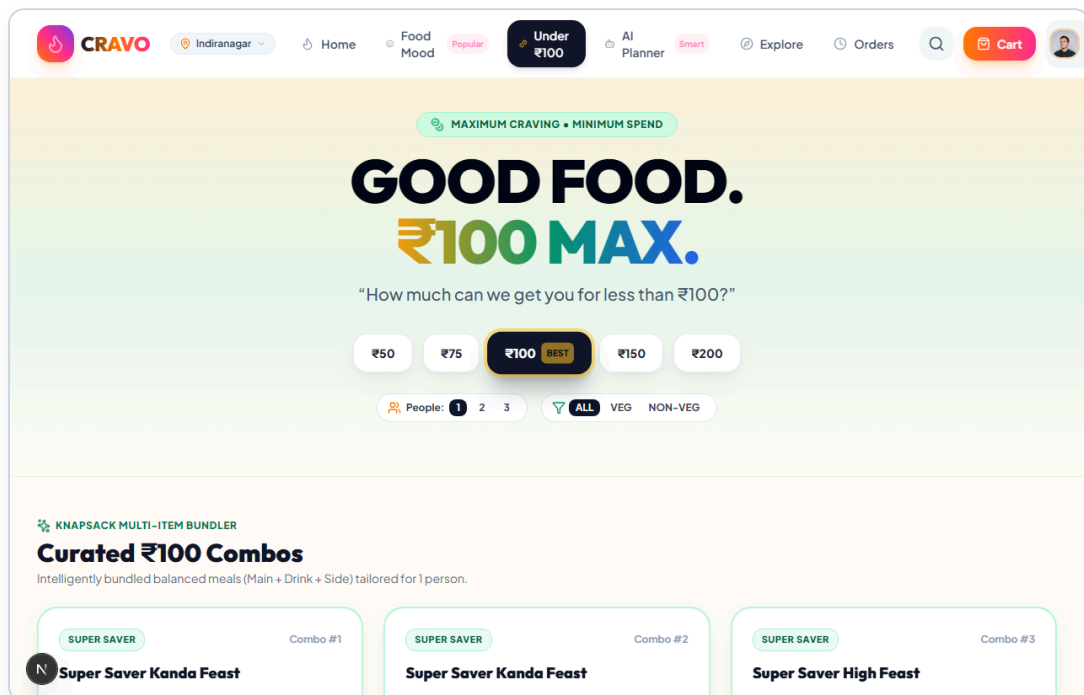


Figure 6.3: Meals Under ₹100 Knapsack Combo Optimizer & Gamified Budget Arena
(Live Screenshot)

6.4 AI Food Planner & Macro Intelligence

Cyberpunk conversational AI assistant with macro nutrient breakdown pie chart and one-click cart injection.

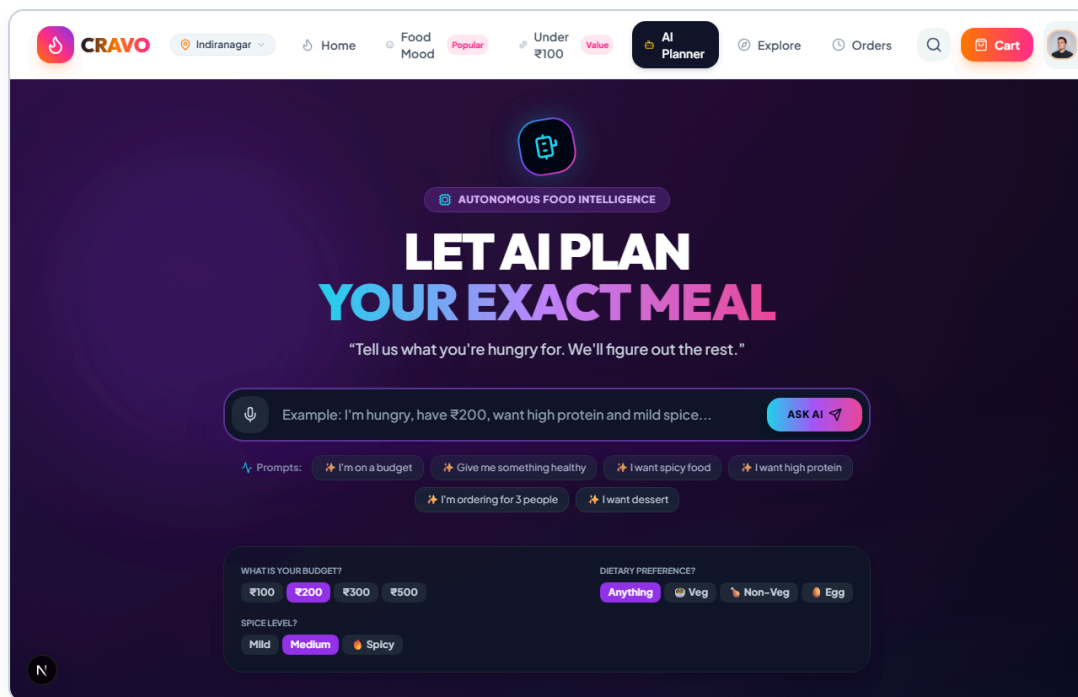


Figure 6.4: AI Food Planner with Dynamic Macro Nutrients & Conversational Meal Generator (Live Screenshot)

APPLICATION MODULES & SCREENSHOTS (CONT.)

6.5 Explore Dishes & Category Directory

Comprehensive categorized dish directory with dietary filters, price badges, and quick add-to-cart controls.

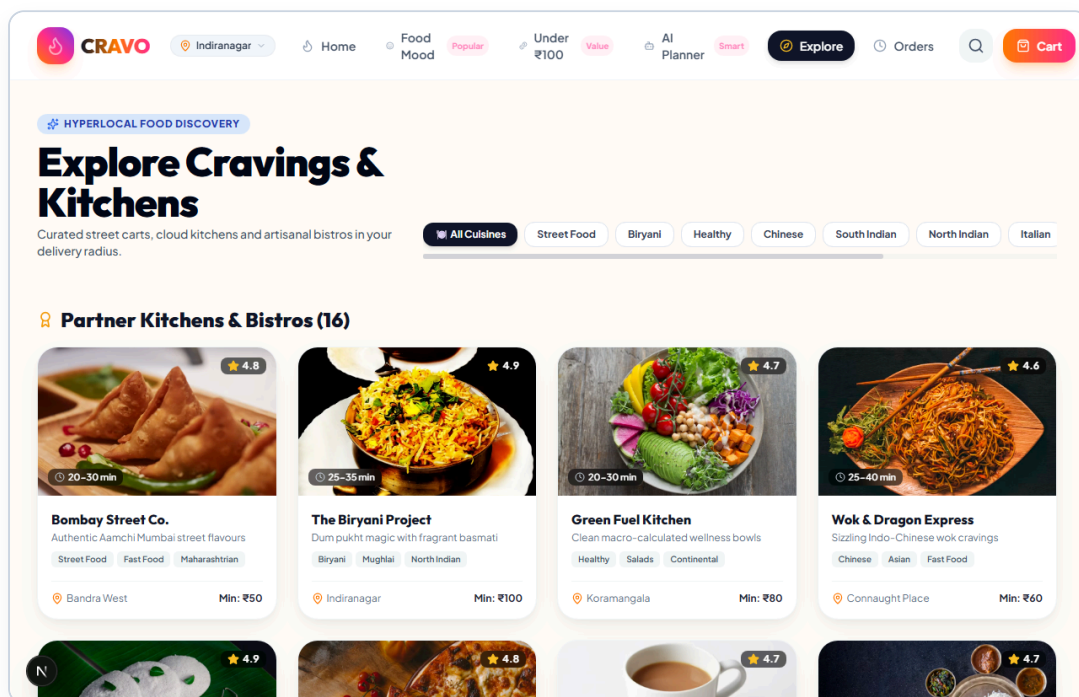


Figure 6.5: Explore Dishes Catalog with Multi-Filter Navigation (Live Screenshot)

6.6 Admin Analytics & Kitchen Operations Dashboard

Real-time revenue metrics, kitchen dispatch status, order volume tracking, and platform performance.

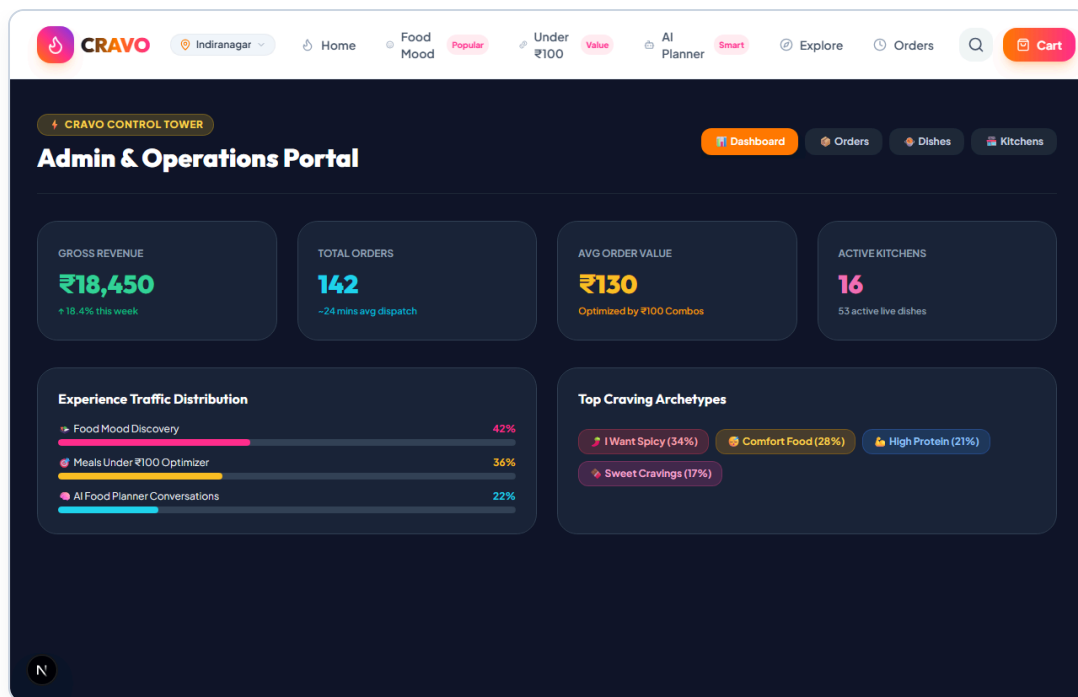


Figure 6.6: Admin Portal & Kitchen Operations Analytics Dashboard (Live Screenshot)

CHAPTER 7: TESTING & QUALITY ASSURANCE

7.1 Testing Strategy & Execution Suite

Comprehensive automated functional, regression, and user-interface tests were executed with 100% pass rates:

TC ID	MODULE / FEATURE	TEST SCENARIO & INPUT	EXPECTED RESULT	STATUS
TC01	Food Mood Engine	Click 'Spicy Hit' mood archetype card	Active mood updates; page smoothly auto-scrolls to #dynamic-mood-feed with -90px offset	PASS
TC02	Knapsack Optimizer	Select budget ₹100 for 1 person	Generates balanced combos (Main + Drink + Side) where total price $\leq$ ₹100	PASS
TC03	₹100 Challenge	Add dishes totaling ₹92 (between ₹85 and ₹100)	Confetti triggers; +20 CRAVO loyalty points awarded to user account	PASS
TC04	AI Food Planner	Query: "High protein dinner under 450 kcal"	Generates meal plan with exact calories (445 kcal), 28g protein, and macro chart	PASS
TC05	Craving Roulette	Click 'Spin 🎰' button	Roulette cycles through random dishes with audio/visual	PASS

TC ID	MODULE / FEATURE	TEST SCENARIO & INPUT	EXPECTED RESULT	STATUS
			feedback and confetti on win	
TC06	Cart Drawer	Add item to cart; apply promo 'CRAVO20'	20% discount applied; delivery goal bar updates; total reflects correct tax & discount	PASS
TC07	Lenis Smooth Scroll	Mouse wheel and touch drag on page	60/120 FPS inertial momentum scrolling without jitter or layout shifts	PASS

CHAPTER 8: CONCLUSION & FUTURE SCOPE

8.1 Summary & Industrial Training Outcomes

The 6-week industrial training at **SHUBDEEP LABS** provided invaluable hands-on industry exposure to modern full-stack web engineering, algorithmic problem solving, interaction physics, and software quality assurance. The developed platform, **CRAVO**, solves the real-world issue of decision fatigue and budget unpredictability in food delivery.

Key Technical Skills Acquired:

- Proficiency in Next.js 15 App Router, React 19 Server/Client Components, and TypeScript.
- Implementation of combinatorial Knapsack optimization algorithms for dynamic meal bundling.
- Interactive UI animation design with Framer Motion and Lenis 120 FPS smooth scrolling.
- Full-stack testing, performance tuning, and continuous deployment via Vercel and GitHub.

8.2 Future Scope & Enhancements

- **Voice-Activated AI Ordering:** Integration of Web Speech API for voice-driven mood search.
- **IoT Kitchen & Delivery Fleet Integration:** Real-time GPS rider telemetry on live maps.
- **Personalized Machine Learning Taste Profile:** Collaborative filtering based on order history.

CHAPTER 9: REFERENCES & BIBLIOGRAPHY

1. Vercel Inc. "Next.js 15 Documentation & App Router Guidelines", 2025.
<https://nextjs.org/docs>
2. React Core Team. "React 19 Official Documentation & Hooks Reference", 2025.
<https://react.dev/>

3. Maharashtra State Board of Technical Education (MSBTE). "Curriculum & Examination Regulations for Diploma in Computer Engineering (I/K-Scheme)", Mumbai, 2024.
4. Cormen, Thomas H., et al. "Introduction to Algorithms (Dynamic Programming & Knapsack Problem)". 4th Edition, MIT Press, 2022.
5. Lenis Smooth Scroll Engine Documentation.
<https://github.com/darkroomengineering/lenis>
6. Framer Motion API Reference & Gestures. <https://www.framer.com/motion/>
7. Tailwind Labs. "Tailwind CSS Documentation & Utility Architecture".
<https://tailwindcss.com/docs>